

Roll No

Total No. of Pages : 02

Total No of Questions : 09

MCA (Sem.-1)

ADVANCED DATABASE MANAGEMENT SYSTEM

Subject Code : PGCA-1953

M.Code : S79038

Date of Examination : 20-01-2025

Time : 3 Hrs.

Max. Marks : 70

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION - B & C. have FOUR questions each.
3. Attempt any FIVE questions from SECTION B & C carrying TEN marks each.
4. Select atleast TWO questions from SECTION - B & C.

SECTION-A

I. Write short notes on :

- a) What is the need of DBMS over traditional Data storage mechanism?
 - b) Define external view and internal view.
 - c) What is meant by Entity set?
 - d) Define Functional Dependency. Give an example.
 - e) Define Relational Algebra.
 - f) Explain XML databases and situations when to use.
 - g) Define the term Inter Query Parallelism.
 - h) What is RDBMS?
 - i) Define Temporal Databases.
10. A student is characterized by his/her Roll No., Name, CGPA. He can take maximum of 5 subjects. A subject can be taken by maximum of 100 students. There are certain subjects which are not taken by any student. Subject is characterized by its id, subject_name, credit and grade.
Represent the cardinality for the above mentioned problem using ER diagram.

SECTION-B

2. Discuss in detail the major components of Database Management System
3. a) What is an Attribute in the Entity-Relationship Model? What are its types?
b) Suppose the following requirements are given for IPL (Indian Premier League)
 - There are many teams and each team has a name, captain, state, set of players, coach. Each player has name, id, age, skill, injury.
 - Each player is associated with only one team. The captain of the team is also a player
 - The game is played between two teams (which are named as team 1 and team 2) in a particular city and with a score.

Draw a clear ER Diagram of IPL to identify the entity sets, attributes and the relationships.

4. Define the term normalization. Explain type of normal forms and dependencies
5. a) Distinguish between Relational Algebra and Relational Calculus.
b) Discuss the term Query Execution and Query Optimization. Why query optimization is required?

SECTION-C

6. Explain XML hierarchical Data Model. Differentiate between data centric, document centric and hybrid XML documents.
7. What do you understand by the concept of distributed database? Discuss various concurrency control protocols in DDBMS.

8. What is Big Data and what are its primary characteristics. Define NoSQL databases.

What are the different types of NoSQL Databases? Discuss open source databases

9. a) What is a multidimensional database and how does it differ from a traditional relational database in terms of data modeling and structure?
b) What are some key benefits of using multidimensional databases in data analytics?

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student